

Oracal Execution

CREATE TABLE

```
CREATE TABLE Testing (RollNo int, FirstName varchar(255), LastName varchar(255),  
MobileNo int, City varchar(255), State varchar(255), Marks int);
```

ALTER TABLE

```
ALTER TABLE Testing
```

```
ADD DOB date;
```

```
ALTER TABLE Testing
```

```
MODIFY DOB TIMESTAMP;
```

```
ALTER TABLE Testing
```

```
DROP COLUMN DOB;
```

INSERT INTO

```
INSERT INTO Testing
```

```
VALUES(201,'MS','Dhoni',98256345895,'Ranchi','Jharkhand',99);
```

```
INSERT INTO Testing
```

```
VALUES(202,'Ruturaj','Gaikwad',75856345895,'Lonawala','Maharashtra',95);
```

```
INSERT INTO Testing
```

```
VALUES (203,'Ravindra','Jadeja',9856231470,'Jamnagar','Rajsathan',91);
```

INSERT INTO Testing

VALUES (204,'Suresh','Raina',9856231212,'Kanpur','Uttar Pradesh',72);

INSERT INTO Testing

VALUES (205,'Deepak','Chahar',9631478520,'Agra','Uttar Pradesh',65);

INSERT INTO Testing

VALUES (206,'Shardul','Thakur',8751478520,'Mumbai','Maharashtra',70);

INSERT INTO Testing

VALUES (207,'Rohit','Sharma',7851478520,'Mumbai','Maharashtra',76);

INSERT INTO Testing

VALUES (208,'KL','Rahul',9841478520,'Bengluru','Karnataka',84);

INSERT INTO Testing

VALUES (209,'Yvraj','Singh',7651478520,'Mohali','Punjab',85);

INSERT INTO Testing

VALUES (210,'Harbhajan','Singh',9151478520,'Mohali','Punjab',85);

INSERT INTO Testing (FirstName,LastName,City,State)

VALUES ('Mangesh','Reddy','Latur','Maharashtra');

SELECT

SELECT*FROM Testing;

SELECT City FROM Testing;

SELECT City,State FROM Testing;

SELECT FirstName,LastName,City,State FROM Testing;

DISTINCT

SELECT DISTINCT City FROM Testing;

SELECT DISTINCT State FROM Testing;

SELECT DISTINCT City,State FROM Testing;

ROWNUM

SELECT*FROM Testing WHERE ROWNUM<=2;

SELECT*FROM Testing WHERE ROWNUM<=5;

AGGREGATE FUNCTION

SELECT COUNT(City) FROM Testing;

SELECT AVG(Marks) FROM Testing;

SELECT SUM(Marks) FROM Testing;

SELECT MAX(Marks) FROM Testing;

SELECT MIN(Marks) FROM Testing;

WHERE CLAUSE

SELECT*FROM Testing

WHERE City='Mohali';

```
SELECT*FROM Testing  
WHERE City='Mumbai';
```

```
SELECT*FROM Testing  
WHERE Marks=85;
```

```
SELECT*FROM Testing  
WHERE Marks>72;
```

```
SELECT*FROM Testing  
WHERE Marks>=72;
```

```
SELECT*FROM Testing  
WHERE Marks<95;
```

```
SELECT*FROM Testing  
WHERE Marks<=95;
```

```
SELECT FirstName,LastName,Marks FROM Testing  
WHERE Marks>=72;
```

```
SELECT*FROM Testing  
WHERE NOT City='Mohali';
```

```
SELECT*FROM Testing  
WHERE NOT City='Mumbai';
```

OR

```
SELECT*FROM Testing  
WHERE City='Pune' OR City='Mumbai';
```

```
SELECT*FROM Testing  
WHERE City='Mohali' OR State='Maharashtra';
```

```
SELECT*FROM Testing  
WHERE FirstName='MS' OR LastName='Singh';
```

```
SELECT City FROM Testing  
WHERE FirstName='MS' OR LastName='Singh';
```

AND

```
SELECT*FROM Testing  
WHERE City='Pune' AND City='Mumbai';
```

```
SELECT*FROM Testing  
WHERE City='Mohali' AND State='Maharashtra';
```

```
SELECT*FROM Testing  
WHERE City='Mohali' AND State='Punjab';
```

```
SELECT*FROM Testing  
WHERE Marks=99 AND RollNo=201;
```

```
SELECT*FROM Testing  
WHERE FirstName='KL' AND LastName='Rahul';
```

```
SELECT State FROM Testing  
WHERE FirstName='KL' AND LastName='Rahul';
```

BETWEEN

```
SELECT*FROM Testing  
WHERE Marks BETWEEN 50 AND 90;
```

```
SELECT*FROM Testing  
WHERE Marks NOT BETWEEN 50 AND 90;
```

```
SELECT*FROM Testing  
WHERE FirstName BETWEEN 'MS' AND 'Yvraj';
```

```
SELECT*FROM Testing  
WHERE FirstName NOT BETWEEN 'MS' AND 'Yvraj';
```

```
SELECT*FROM Testing
```

WHERE RollNo BETWEEN 201 AND 205;

IN

SELECT*FROM Testing

WHERE City IN('Mumbai','Mohali','Lonawala');

SELECT*FROM Testing

WHERE City NOT IN('Mumbai','Mohali','Lonawala');

SELECT*FROM Testing

WHERE State NOT IN('Panjab','Maharashtra');

SELECT*FROM Testing

WHERE Marks IN(85);

SELECT*FROM Testing

WHERE Marks NOT IN(85);

SELECT*FROM Testing

WHERE FirstName NOT IN('KL');

ORDER BY

SELECT*FROM Testing

ORDER BY Marks;

```
SELECT*FROM Testing
```

```
ORDER BY FirstName;
```

```
SELECT*FROM Testing
```

```
ORDER BY LastName;
```

```
SELECT*FROM Testing
```

```
ORDER BY FirstName,Marks;
```

```
SELECT*FROM Testing
```

```
ORDER BY Marks DESC;
```

```
SELECT*FROM Testing
```

```
ORDER BY FirstName DESC;
```

```
SELECT*FROM Testing
```

```
ORDER BY LastName DESC;
```

ALIAS

```
SELECT MobileNo AS MbNo FROM Testing;
```

```
SELECT FirstName AS FN, LastName AS LN FROM Testing;
```

UPDATE

UPDATE Testing

SET MobileNo=8888888888

WHERE FirstName='MS';

UPDATE Testing

SET MobileNo=9999999999

WHERE RollNo=202;

UPDATE Testing

SET MobileNo=9999999999, Marks=68

WHERE RollNo=202;

UPDATE Testing

SET City='Pune'

WHERE City='Lonawala';

DELETE

DELETE FROM Testing

WHERE FirstName='Harbhajan';

Note- Click on the Rollback or Press F12 key

Deleted records again back in table

LIKE

```
SELECT*FROM Testing  
WHERE FirstName Like 'R%';
```

```
SELECT*FROM Testing  
WHERE LastName Like '%r';
```

```
SELECT*FROM Testing  
WHERE LastName Like '%ak%';
```

```
SELECT*FROM Testing  
WHERE FirstName Like 'R%j';
```

```
SELECT*FROM Testing  
WHERE FirstName Like 'R%t';
```

WALIDCARD

```
SELECT*FROM Testing  
WHERE FirstName Like '_S';
```

```
SELECT*FROM Testing  
WHERE FirstName Like '___hit';
```

```
SELECT*FROM Testing  
WHERE FirstName Like '_oh__';
```

```
SELECT*FROM Testing  
WHERE FirstName Like 'R_____';
```

```
SELECT*FROM Testing  
WHERE FirstName Like 'Ro_____';
```

UNION

```
CREATE TABLE Team (RollNo int, FirstName varchar(255), LastName varchar(255),  
MobileNo int, City varchar(275), Marks int, State varchar(255));
```

```
INSERT INTO Team  
VALUES(111, 'Rohit', 'Sharma', 9623547854, 'Mumbai', 78, 'Maharashtra');
```

```
INSERT INTO Team  
VALUES(101, 'Shubham', 'Gill', 9623787894, 'Pune', 50, 'Maharashtra');
```

```
INSERT INTO Team  
VALUES(102, 'Pruthvi', 'Shaw', 7723787895, 'Pune', 49, 'Maharashtra');
```

```
INSERT INTO Team  
VALUES(103, 'Hanuma', 'Vihari', 9737878954, 'Mohali', 48, 'Panjab');
```

```
INSERT INTO Team  
VALUES(104, 'Suryakumar', 'Yadhav', 9823778954, 'Nagpur', 45, 'Maharashtra');
```

INSERT INTO Team

VALUES(107, 'Virat', 'Kholi', 7777878954, 'Delhi', 49, 'Delhi');

INSERT INTO Team

VALUES(108, 'MS', 'Dhoni', 9600878954, 'Ranchi', 85, 'Jharkhand');

INSERT INTO Team

VALUES(109, 'Yuvraj', 'Singh', 77237878954, 'Mohali', 85, 'Panjab');

INSERT INTO Team

VALUES(110, 'KL', 'Rahul', 98237878954, 'Bengluru', 95, 'Karnataka');

SELECT LastName FROM Testing

UNION

SELECT LastName FROM Team;

SELECT Marks FROM Testing

UNION

SELECT Marks FROM Team;

SELECT LastName FROM Testing

UNION ALL

SELECT LastName FROM Team;

```
SELECT Marks FROM Testing
```

```
UNION ALL
```

```
SELECT Marks FROM Team;
```

TRUNCATE TABLE

```
TRUNCATE TABLE Testing;
```

DROP TABLE

```
DROP TABLE Testing;
```

JOIN

```
CREATE TABLE Client(ID int,Name varchar(255),Age int,Address varchar(255),Salary int);
```

```
INSERT INTO Client VALUES(1,'Ramesh',32,'Ahmedabad',2000);
```

```
INSERT INTO Client VALUES(2,'Khilan',25,'Delhi',1500);
```

```
INSERT INTO Client VALUES(3,'kaushik',23,'Kota',2000);
```

```
INSERT INTO Client VALUES(4,'Chaitali',25,'Mumbai',6500);
```

```
INSERT INTO Client VALUES(5,'Hardik',27,'Bhopal',8500);
```

```
INSERT INTO Client VALUES(6,'Komal',22,'MP',4500);
```

```
INSERT INTO Client VALUES(7,'Muffy',24,'Indore',10000);
```

```
CREATE TABLE Buyer (OID int, MobileNo int, ID int, Amount int);
```

```
INSERT INTO Buyer VALUES(102,8888123654,3,3000);
```

```
INSERT INTO Buyer VALUES(100,9999123654,3,1500);
```

```
INSERT INTO Buyer VALUES(101,7777123654,2,1560);
```

```
INSERT INTO Buyer VALUES(103,8989123654,4,2060);
```

INNER JOIN

```
SELECT Name,Address,Amount FROM
```

```
Client INNER JOIN Buyer
```

```
ON Client.ID=Buyer.ID;
```

```
SELECT * FROM
```

```
Client INNER JOIN Buyer
```

```
ON Client.ID=Buyer.ID;
```

LEFT JOIN

```
SELECT Name,Address,Amount FROM
```

```
Client LEFT JOIN Buyer
```

```
ON Client.ID=Buyer.ID;
```

```
SELECT * FROM
```

```
Client LEFT JOIN Buyer
```

```
ON Client.ID=Buyer.ID;
```

RIGHT JOIN

```
SELECT Name,Address,Amount FROM
```

```
Client RIGHT JOIN Buyer
```

```
ON Client.ID=Buyer.ID;
```

```
SELECT * FROM  
Client RIGHT JOIN Buyer  
ON Client.ID=Buyer.ID;
```

FULL JOIN

```
SELECT Name,Address,Amount FROM  
Client FULL JOIN Buyer  
ON Client.ID=Buyer.ID;
```

```
SELECT * FROM  
Client FULL JOIN Buyer  
ON Client.ID=Buyer.ID;
```

GROUP BY

```
SELECT MAX(Salary) FROM Client  
GROUP BY Name;
```

```
SELECT Name, MAX(Salary) FROM Client  
GROUP BY Name;
```

```
SELECT MIN(Salary) FROM Client  
GROUP BY Name;
```

```
SELECT Name, MIN(Salary) FROM Client  
GROUP BY Name;
```

```
SELECT SUM(Salary) FROM Client
```

```
GROUP BY Name;
```

```
SELECT Name, SUM(Salary) FROM Client
```

```
GROUP BY Name;
```

HAVING

```
SELECT SUM(Salary) FROM Client
```

```
GROUP BY Name
```

```
HAVING SUM(Salary)<=4500;
```

```
SELECT Name, SUM(Salary) FROM Client
```

```
GROUP BY Name
```

```
HAVING SUM(Salary)<=4500;
```

```
SELECT SUM(Salary) FROM Client
```

```
GROUP BY Name
```

```
HAVING SUM(Salary)>=4500;
```

```
SELECT Name, SUM(Salary) FROM Client
```

```
GROUP BY Name
```

```
HAVING SUM(Salary)>=4500;
```

Find Salary

```
CREATE TABLE Bollywood (FirstName varchar(255),Salary int);
```

```
INSERT INTO Bollywood VALUES ('Emraan',50000);
```

```
INSERT INTO Bollywood VALUES ('John',60000);
```

```
INSERT INTO Bollywood VALUES ('Ritesh',40000);
```

```
INSERT INTO Bollywood VALUES ('Akshay',90000);
```

```
INSERT INTO Bollywood VALUES ('Shahrukh',45000);
```

```
INSERT INTO Bollywood VALUES ('Deepika',40000);
```

```
INSERT INTO Bollywood VALUES ('Shraddha',35000);
```

```
INSERT INTO Bollywood VALUES ('Salman',65000);
```

```
INSERT INTO Bollywood VALUES ('Ayushman',56000);
```

```
INSERT INTO Bollywood VALUES ('Abhishek',76000);
```

```
INSERT INTO Bollywood VALUES ('kriti',59000);
```

```
INSERT INTO Bollywood VALUES ('kirti',54000);
```

```
INSERT INTO Bollywood VALUES ('Rajkumar',52000);
```

```
INSERT INTO Bollywood VALUES ('Karina',65000);
```

```
INSERT INTO Bollywood VALUES ('Katrina',71000);
```

```
INSERT INTO Bollywood VALUES ('Ajay',92000);
```

1st Maximum Salary

```
SELECT MAX(Salary) FROM Bollywood;
```

1st Minimum Salary

```
SELECT MIN(Salary) FROM Bollywood;
```

2nd Maximum Salary

```
SELECT MAX(Salary) FROM Bollywood  
WHERE Salary NOT IN (SELECT MAX(Salary) FROM Bollywood);
```

2nd Minimum Salary

```
SELECT MIN(Salary) FROM Bollywood  
WHERE Salary NOT IN (SELECT MIN(Salary) FROM Bollywood);
```

Nth Maximum / Highest Salary

```
SELECT * FROM ( SELECT * FROM ( SELECT * FROM ( SELECT * FROM Bollywood  
ORDER BY Salary DESC) WHERE ROWNUM<=N) ORDER BY Salary) WHERE  
ROWNUM<=1;
```

3rd Highest Salary

```
SELECT * FROM ( SELECT * FROM ( SELECT * FROM ( SELECT * FROM Bollywood  
ORDER BY Salary DESC) WHERE ROWNUM<=3) ORDER BY Salary) WHERE  
ROWNUM<=1;
```

5th Highest Salary

```
SELECT * FROM ( SELECT * FROM ( SELECT * FROM ( SELECT * FROM Bollywood  
ORDER BY Salary DESC) WHERE ROWNUM<=5) ORDER BY Salary) WHERE  
ROWNUM<=1;
```

7th Highest Salary

```
SELECT * FROM ( SELECT * FROM ( SELECT * FROM ( SELECT * FROM Bollywood  
ORDER BY Salary DESC) WHERE ROWNUM<=7) ORDER BY Salary) WHERE  
ROWNUM<=1;
```

Nth Minimum / Lowest Salary

```
SELECT * FROM ( SELECT * FROM ( SELECT * FROM ( SELECT * FROM Bollywood  
ORDER BY Salary ) WHERE ROWNUM<=N) ORDER BY Salary DESC) WHERE  
ROWNUM<=1;
```

3rd Lowest Salary

```
SELECT * FROM ( SELECT * FROM ( SELECT * FROM ( SELECT * FROM Bollywood  
ORDER BY Salary ) WHERE ROWNUM<=3) ORDER BY Salary DESC) WHERE  
ROWNUM<=1;
```

5th Lowest Salary

```
SELECT * FROM ( SELECT * FROM ( SELECT * FROM ( SELECT * FROM Bollywood  
ORDER BY Salary ) WHERE ROWNUM<=5) ORDER BY Salary DESC) WHERE  
ROWNUM<=1;
```

7^h Lowest Salary

```
SELECT * FROM ( SELECT * FROM ( SELECT * FROM ( SELECT * FROM Bollywood  
ORDER BY Salary ) WHERE ROWNUM<=7) ORDER BY Salary DESC) WHERE  
ROWNUM<=1;
```

Constraints

```
CREATE TABLE TeamIndia (RollNo int Primary Key, FirstName varchar(255)NOT NULL,  
LastName varchar(255), MobileNo int Unique, City varchar(255) Default 'Pune', State  
varchar(255), marks int);
```